

LocalRCA: On-Device Root Cause Analysis for Windows Endpoints Without Ground Truth

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Abstract—Diagnosing why a single Windows machine slowed down or failed is poorly served by fleet observability tools, which assume a network, a backend and consent to export telemetry. We present LocalRCA, a desktop system that records telemetry without sending any of it off the machine, learns that machine’s normal behaviour with an LSTM autoencoder, and explains anomalies with Granger causality constrained by false-discovery-rate control, an effect-size floor and a subsystem topology prior. A personal machine has no labelled incidents, so we evaluate with seven controlled fault injections and a replay over 596 incidents found in real collected history. The system detects and correctly attributes injected CPU, disk and memory faults, and in one case explained a fault it was never told about. On real history it produces a supported causal chain for 106 of 596 incidents (17.8%, 95% CI 14.9–21.1%); 36% are too short to test. When it does answer, its top cause differs from the most anomalous metric about half the time. The hand-written subsystem prior rejects 22.7% of statistically accepted pairs, including mechanisms that demonstrably exist. We argue that the most useful property of such a tool is saying when it cannot explain an incident.

Index Terms—root cause analysis, anomaly detection, LSTM autoencoder, Granger causality, endpoint telemetry, false discovery rate, explainability

I. INTRODUCTION

Diagnosing why a personal computer became slow, stalled or crashed is typically a manual exercise: an engineer lines up resource graphs against the Windows Event Log and guesses. Fleet-oriented observability platforms automate this, but they assume infrastructure a single machine does not have: a collector network, a time-series backend, and the user’s consent to export telemetry that shows which applications they run.

Targeting a single endpoint imposes three constraints, and most of the design follows from them.

No egress. Telemetry about application usage is sensitive, so the system transmits nothing. The collector contains no network code, which means privacy does not depend on a setting that someone could change.

No ground truth. A personal machine has no labelled incidents. The model cannot be trained to recognise known faults and cannot be evaluated against a labelled test set. This constraint shaped the evaluation more than any other. The only workable response we found was to manufacture ground truth by causing faults on purpose (Section V).

Cold start is unavoidable. “Normal” has to be learned from the machine itself, so the tool can do nothing until it has

seen enough normal behaviour. In our configuration that takes about 10.5 hours of clean collection.

Our contributions are:

- 1) An end-to-end on-device RCA pipeline combining reconstruction-error anomaly detection with statistically constrained Granger causality (Section IV).
- 2) A fault-injection harness that manufactures ground truth on a machine that has none, and a demonstration that the pipeline can identify a cause it was not told about (Section V).
- 3) A population-scale measurement of how often the causal layer produces an answer at all, replicated on a second survey, and of how often that answer differs from simply naming the most anomalous metric (Section VI).
- 4) An audit showing that the hand-written subsystem prior removes more statistically accepted pairs than any later stage of the pipeline, and that it forbids physically real mechanisms (Section VII).
- 5) A deployment design for a tool that holds sensitive telemetry (per-user encryption of the learned model, signed binaries, and an interface that separates the verdict from the evidence), including where it falls short (Section VIII).

II. RELATED WORK

Anomaly detection in multivariate time series. Surveys of the field [1], [2] separate detectors by whether labels exist and by whether the anomaly is a point, a subsequence or a whole series. Reconstruction-based detectors built on LSTMs [3] are a standard approach when labelled anomalies are unavailable [4], [5]. They train on normal data only and threshold the reconstruction error, so they need no labelled faults, which an endpoint does not have. Later detectors improve accuracy with stochastic recurrent models (OmniAnomaly [6]), adversarially trained autoencoders (USAD [7]), signature matrices fed to convolutional LSTMs (MSCRED [8]), learned sensor graphs with attention (GDN [9]) and transformers (TranAD [10]). MSCRED and GDN also point to the sensors responsible for an anomaly; none of them says why those sensors moved. We keep the plain LSTM autoencoder because its per-metric reconstruction error maps directly onto the metric to be explained and it trains on a commodity CPU. Whether the heavier detectors would pay for themselves on a single endpoint is not tested here.

Log-based anomaly detection. System logs are the other common signal. Xu et al. [11] applied PCA to message-count vectors mined from console logs, He et al. [12] benchmarked six supervised and unsupervised log detectors, and DeepLog [13] learns log-key sequences with an LSTM and flags keys it did not predict. On a personal machine the Windows Event Log is sparse (our host logged 6,600 allow-listed records in 59 days), so LocalRCA uses it only as context around a metric anomaly and not as a detector.

Causal discovery from observational data. Granger causality [14] tests whether one series improves prediction of another beyond the second’s own history. It is correlational under a temporal constraint rather than interventional, and is well known to over-report on correlated series without correction. We apply Benjamini–Hochberg false-discovery-rate control [15] across the full set of ordered pairs, an effect-size floor, and a domain prior over subsystems. Nonlinear extensions replace the linear autoregression with neural networks whose sparse input weights identify Granger causes [16], and PCMCi [17] conditions on each series’ own past and a selected parent set to suppress false links caused by autocorrelation. Both need longer series than an incident window usually provides: Section VI shows that 36% of real incidents are too short even for the linear test.

Root cause analysis in distributed systems. Microservice RCA work typically exploits service call graphs or distributed traces to constrain candidate causes [18], [19], and Soldani and Brogi [20] review how faults are found and traced in cloud microservice deployments. Most methods first work out which component depends on which and then traverse that structure. CloudRanger [21] infers causal structure through PC-based discovery and ranks candidates by a second-order random walk; AutoMAP [22] builds behaviour graphs over several metrics; MicroRCA [23] runs personalised PageRank over an attributed service–host graph; and MicroCause [24] pairs a PCMCi-derived causal graph with a temporal cause-oriented random walk. CIRCA [25] derives a structural graph from the system architecture and treats root-cause identification as recognising which variable was intervened on, while BARO [26] needs no graph and ranks each metric by how sharply its distribution shifts, estimated with a Bayesian change-point model. A single endpoint has no service topology, so we write an explicit subsystem map by hand, and Section VII shows how much the quality of that map then matters.

Benchmarked causal RCA at fleet scale. Recent work evaluates causal discovery for root cause analysis at a scale this system does not attempt. Ikram et al. [27] discover root causes across thousands of microservice nodes and report top- k recall against ground truth gathered from a real production incident corpus. Pham et al. [28] benchmark six causal-RCA methods (including CausalRCA and BARO) across microservice architectures and find that none dominates uniformly and that results depend heavily on the call graph. Both assume a call graph or trace corpus that a single endpoint does not have. Our work differs in running on one machine, with no ground truth,

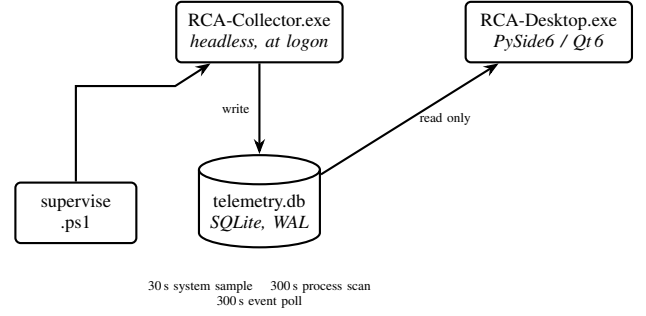


Fig. 1. Process architecture. The collector writes; the desktop application only reads. A PowerShell supervisor restarts the collector if it exits unexpectedly, because creating a Task Scheduler logon task requires elevation that a diagnostic tool should not demand.

and in reporting when it cannot explain an incident instead of ranking whatever the evidence.

III. SYSTEM ARCHITECTURE

The system is two processes that share a SQLite database (Fig. 1). They are separate because collection has to keep running after the user closes the interface.

The collector records 29 channels every 30 seconds, the top processes by CPU and resident memory every 300 seconds (tightening to 30 seconds under load), and an allow-listed subset of Windows Event Log records. Retention is tiered by sensitivity: numeric readings for 365 days, per-process detail for 30 days, and the foreground application name, the most personal field collected, for 30 days.

The evaluation host is a laptop with an Intel Core i7-14700HX (28 logical processors) and 16 GB of RAM, running Windows 11 Home. At the time of the first survey it had accumulated 124,674 system samples, 1,835,250 process samples and 6,600 Event Log records over 59 days, occupying 227 MB. Coverage of that span is 73.1%: 157 breaks in collection account for 383 hours not recorded. The population survey (Section VI) draws on the most recent 30 days of this history.

Collection is cheap but not free. Measured over 5- and 15-minute windows, the collector used 14–21% of one logical processor (0.5–0.7% of the host’s capacity) with a 16 MB working set. Across 133,753 samples the spacing had a median of 30.0 s and a 99th percentile of 30.005 s, so the cost does not delay sampling.

IV. METHOD

A. Scaling and windowing

A min–max scaler is fitted on the clean baseline only, $x'_j = (x_j - \min_j) / (\max_j - \min_j)$, and incident windows are transformed with the baseline’s parameters and clipped to $[0, 1]$. Fitting on the incident would normalise away the deviation being sought.

Training windows are built *within contiguous segments only*, never spanning a collection gap. A segment is a maximal run of samples whose spacing is within tolerance of the sampling cadence.

B. Anomaly detection

An LSTM autoencoder compresses a window of $T = 12$ samples (six minutes) over $n = 17$ features to a 32-dimensional latent vector and reconstructs it. The 17 are the numeric resource metrics the host reports; the other recorded channels are raw counters, GPU and idle readings, the foreground application, and two battery fields that are empty on this host, and are not modelled. Only the encoder's final hidden state reaches the bottleneck, so the entire window is represented by 32 numbers before reconstruction; that compression is the detection mechanism. Encoder and decoder are two-layer LSTMs with 64 hidden units and dropout 0.2 (126,385 parameters). Training minimises

$$\mathcal{L} = \frac{1}{BTn} \sum_{b,t,j} (x_{btj} - \hat{x}_{btj})^2. \quad (1)$$

Error is reduced over time but *not* over features, which is what makes the output attributable to a metric rather than to a window:

$$s_j = \frac{1}{T} \sum_{t=1}^T (x_{tj} - \hat{x}_{tj})^2. \quad (2)$$

We train with Adam (learning rate 10^{-3} , batch 32) for five epochs on windows taken at stride 5 within clean segments, hold out the last 20% of windows in time order for validation, and keep the epoch with the lowest validation loss. Training starts once 250 windows exist, which is the 10.5 hours quoted above. Per-metric thresholds are the 99th percentile of validation error, $\tau_j = P_{99}\{s_j(w)\}$, and a metric is anomalous when $s_j > \tau_j$. Per-metric thresholds are necessary because reconstructability varies widely; a near-constant channel reconstructs almost perfectly, while a bursty one does not, and a single global threshold would flag the latter permanently. By construction about 1% of normal windows exceed each threshold, so an incident needs at least three consecutive flagged samples.

C. Constrained causal inference

For each ordered pair of anomalous metrics, both series are differenced until an Augmented Dickey–Fuller test [29] rejects a unit root (at most twice), then tested for Granger causality at lags $1 \dots L$. The lag minimising the p -value is selected. A pair is skipped unless

$$N \geq 3L + 2, \quad (3)$$

which at the default $L = 5$ requires 17 aligned samples. Most of this system's negative results come from Equation (3).

Surviving p -values are corrected with Benjamini–Hochberg: sorting ascending, the largest rank k satisfying $p_{(k)} \leq \alpha k/m$ is found and ranks $1 \dots k$ accepted, with $\alpha = 0.05$. A p -value alone is not enough, because with a long history even a negligible improvement becomes significant, so we also require an effect-size proxy:

$$\text{effect} = \frac{F}{F + N} \geq 0.10. \quad (4)$$

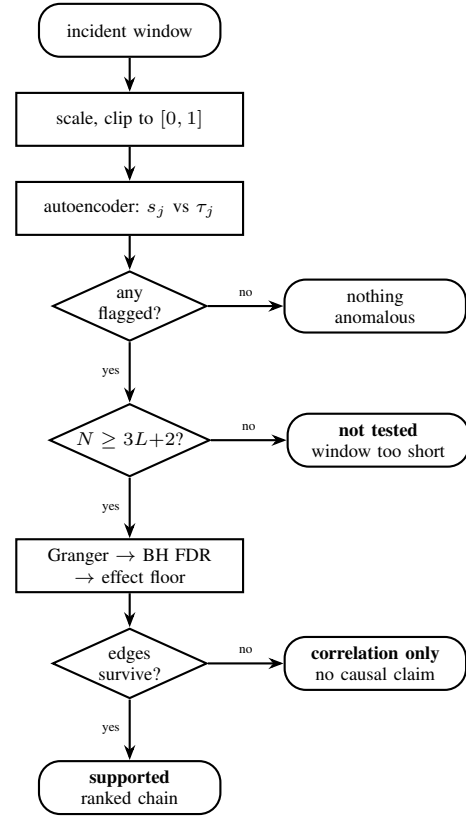


Fig. 2. Inference pipeline. Four terminal states are reported distinctly. Collapsing “not tested” into “nothing survived” would conflate the absence of a result with a negative result, which earlier revisions of this system did.

This quantity is monotone in F , lies in $(0, 1)$, and shrinks as N grows. That is the correction we want, since a large N is what made the p -value small.

Accepted pairs form a directed graph; cycles are removed preferring edges that contradict temporal precedence, and remaining edges are filtered against a subsystem topology prior (Section VII). Candidates are then ranked by a weighted composite plus PageRank [30] on the reversed graph:

$$\text{score}(v) = \min\left(1, 0.70 \sum_k w_k s_k(v) + 0.30 \text{PR}(v)\right), \quad (5)$$

with weights 0.40 causal outflow, 0.30 temporal priority, 0.20 low inflow, 0.05 severity and 0.05 event correlation.

A structural caveat. With zero surviving edges, every node has outflow 0 and inflow 1, so 60% of the weight becomes constant and the ranking collapses onto severity and timing while still emitting a score. The system therefore reports the distinction explicitly rather than presenting the ranking as causal.

V. EVALUATION BY FAULT INJECTION

Because a personal machine provides no labelled incidents, we manufacture ground truth: a harness causes a specific, known disturbance, waits for samples to land at the real cadence, runs the production pipeline over the injection window, and scores the result.

TABLE I
CONTROLLED FAULT INJECTIONS, 30-MINUTE WINDOWS UNLESS NOTED.
FLAGGED COUNTS ARE OUT OF THE 17 MODELLED METRICS.

Fault	Samples	Flagged	Edges	Outcome
CPU, 7 min	14	6/17	—	not tested (below floor)
CPU	60	6/17	6	explained correctly
Disk	60	4/17	0	detected, unexplained
Memory	60	2/17	0	wrong process named
Memory (refixed)	60	3/17	0	detected, attributed
Memory (clean)	60	0/17	—	not measurable on host
Idle	60	1/17	—	5.9% false positives

A. A correct explanation of an unknown cause

Half the CPU cores were burned for 30 minutes with no information given to the pipeline. The ranking named `cpu_pct` first at 1.000, and the surviving edges point *away* from CPU. That is what a real root cause should look like, since nothing upstream drives it:

`cpu_pct_max_core` $\rightarrow$ `disk_busy_pct`
(lag 2, 0.756).

B. Short windows starve the causal layer

The same fault at 7 and at 30 minutes differed only in duration. At 14 samples, against the floor of 17 from (3), *no pair was compared at all*, and the system reported exactly that. At 60 samples, ten pairs survived correction and six survived the topology filter. The causal layer worked; it was short of samples, so the fix is a wider window rather than a redesign. Section VI confirms this at scale.

C. A defect that injection exposed

The memory injection detected the anomaly and then attributed it to four unrelated processes. Attribution ranked candidates by mean CPU, while resident memory was selected but never sorted on, so a process that allocates memory and then sleeps could never be named. A memory-bound cause was unreportable *by construction* in every incident the system had ever analysed. The CPU and disk faults are both CPU-heavy, so they happened to be attributed correctly, and that hid the defect.

D. A measurement this host cannot provide

A later memory injection on a deliberately cleared machine flagged nothing. The detector was correct: the host’s median memory utilisation across 30,199 samples is 95.2%, and the injection window sat at 87.4%, the 9th percentile of the host’s own history. For safety the injection is limited to half of free memory, and on a host that is usually full that cannot push usage above the level the machine already reaches on its own. The fault is undetectable *by construction* here, and the harness now reports this as inconclusive rather than as a detector failure.

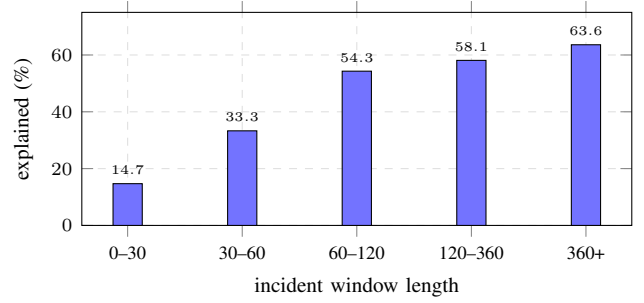


Fig. 3. Causal yield against window length, over 381 analysable incidents ($L = 5$, 30-day survey). The rate rises steadily with window length, which is the short-window effect first seen in a pair of injections, now at scale.

VI. POPULATION-SCALE EVALUATION

Fault injection establishes correctness on a handful of cases but yields single observations. To measure *frequency*, the pipeline was replayed over every incident found in real collected history, read-only and with no injection.

Of 596 incidents discovered, 215 (36%) fall below the sample floor of (3) and cannot be tested at any setting. Of the 381 that can be, 106 produce a supported causal chain. The end-to-end figure a user experiences is therefore $106/596 = 17.8\%$ (95% Wilson interval 14.9–21.1%): roughly one incident in six receives a causal explanation.

We report the whole funnel. Counting only *analysable* incidents would give 27.8% (23.6–32.5%), which looks better but describes fewer situations.

The floor is a parameter artefact. Short incidents are widened before analysis to exactly the model’s window size, 12 samples (5.5 minutes), while (3) at $L = 5$ requires 17. The two constants were chosen independently in the same pipeline. Re-running the survey at $L \in \{3, 4, 5\}$ over an identical population (598 incidents discovered over 30 days of real collected history) confirms the cliff disappears at $L \leq 3$: 0 incidents fall below the floor, versus 215 at $L = 5$. Explanations rise only modestly, from 106 to 110, while the tested population nearly triples (381 to 598), and median surviving edges holds at 2 throughout. What improves is the reporting: more incidents read “tested, nothing found” instead of “not tested”, but few gain an explanation.

Replication. Four days later we re-ran the survey at $L = 5$ on the then most recent 30 days. The two windows share about 26 days, so this checks stability rather than giving an independent sample. It found 520 incidents, of which 177 (34.0%) fell below the floor and 94 were explained (18.1%, 15.0–21.6%), in line with the first run.

Does causality change the answer? A ranking that always named the most anomalous metric would need no causal layer. For each of the 94 explained incidents we compared the top-ranked cause with the metric that had the largest anomaly score. They agree in 46 (48.9%, 39.1–58.9%) and fall in the same subsystem in 62 (66.0%). Where they differ, the causal ranking most often puts a disk metric ahead of a pressure symptom, for example `disk_free_pct` ahead of

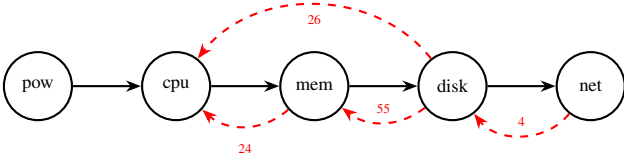


Fig. 4. The subsystem prior is a strict total order (solid), so *network* is a sink and nothing can be reported as caused by network activity. Dashed edges are directions the statistics accepted and the map rejected, labelled with occurrence counts (30-day, 596-incident survey). Two of them describe real mechanisms: the file cache consuming memory under write pressure, and interrupt handling costing CPU during disk activity; a download writing to disk (*network*→*disk*) remains a smaller but real third case.

`swap_used_delta`. This shows the causal ranking is not severity under another name. It does not show the changed answer is the right one, which needs incidents with a known cause.

VII. AUDITING THE SUBSYSTEM PRIOR

Since no service call graph exists on an endpoint, the system encodes domain knowledge as an explicit map of permissible subsystem influence. Auditing it across all analysable incidents shows it is the largest single filter in the pipeline after the statistical gates: of 538 accepted pairs, 337 (62.6%) survive, 122 (22.7%) are rejected by the map and 79 (14.7%) by cycle-breaking.

Every rejected transition has *zero* survivors, because the map forbids whole classes rather than individual edges (Fig. 4). Judged individually, *disk*→*memory* is almost certainly real (the Windows standby file cache grows under write pressure) and is the strongest single rejected relationship measured in this run, at 0.654 (*disk_write_bps*→*swap_used_delta*, lag 2). *disk*→*cpu* reflects interrupt and DPC handling, and *network*→*disk* is the download-writes-to-disk case argued in the prior revision of this analysis, present but smaller in this run (4 occurrences). Conversely, the smallest rejected classes (*network*→*process*, *disk*→*process*, at 1–2 occurrences each) are plausibly spurious rather than real mechanisms being wrongly forbidden.

The prior only runs one way: it models pressure flowing downhill and cannot express the feedback that makes I/O expensive. Adding reverse edges would make it cyclic, and cycle-breaking already removes 14.7% of pairs, so we leave this as an open question rather than patch it.

VIII. DEPLOYMENT ENGINEERING

A research prototype can leave its data in plaintext and show its results as raw tables. A tool that someone else installs, and that records when and how they use their machine, should not. This section describes what the released application (v1.5.2) does about both problems and where it falls short.

A. Protecting the model at rest

The trained model holds per-metric bounds and thresholds, a compact profile of how the machine is used. It is wrapped

with the Windows Data Protection API (DPAPI), which binds it to the user’s logon, and written atomically. The database and logs rely on the Encrypting File System (EFS), which does not exist on Windows Home. The evaluation host runs Home, so its 227MB database is in plaintext, protected only by file permissions, and its model, trained before encryption was added, stays plaintext until retrained. Review also found that an interrupted training run left a plaintext checkpoint on disk; it is now removed in a `finally` block.

B. Signed binaries

Both executables are Authenticode-signed and timestamped, and each release publishes a SHA-256 checksum. The certificate is self-signed, so it proves a binary is unchanged but not who built it, and Windows SmartScreen still warns on first run.

C. Separating the verdict from the evidence

The first screen used to show every captured channel with its raw value, and stated training readiness four different ways. Hiding this would contradict the project’s premise, since a user cannot consent to collection they cannot inspect.

The released interface shows the same computation two ways, chosen by one switch that is off by default. The simple view opens each tab with a single coloured sentence (for example, *Healthy* only at $\geq 90\%$ coverage with at most three breaks) and opens inference on the plain-language report. The advanced view adds the raw counts, channel table, hyperparameters, lag and score columns. Both views read the same figures; only presentation changes.

This design has been checked by one expert review, not with users.

IX. DISCUSSION AND THREATS TO VALIDITY

All measurements come from one machine. Nothing here shows that the results generalise. One measurement, memory-fault detection, cannot be made on this host at all, so testing on a second machine is required, not optional.

Correctness is not established at population scale. The survey’s 596 incidents have no known cause, so the 17.8% explanation rate says how often the causal layer gives an answer, not whether the answer is right. It is an upper bound on usefulness rather than a measure of accuracy. For the same reason we report no precision, recall or F1 for the detector: seven injected windows show that detection works, not how often.

No fault has been tested whose cause is not also the loudest metric. Both explaining runs put the injected fault atop a severity ranking, so a correct causal answer and a correct severity answer remain indistinguishable. On real incidents the two rankings differ about half the time (Section VI), but none of those incidents has a known cause.

Granger causality is not causation. It is correlation under a temporal constraint. The FDR correction, effect floor and subsystem prior exist to suppress the resulting over-reporting, and Section VII shows the prior itself is imperfect.

A methodological observation. Nine defects were found in code with a passing suite of 130 tests, each because a test checked something next to what mattered (a database value instead of the bytes on disk, a function instead of its caller). Adversarial review, not more tests, caught them, and it caught the plaintext checkpoint above the same way. The suite now has 195 tests.

X. CONCLUSION

We presented an on-device root cause analysis system for Windows endpoints operating without ground truth, network access or fleet context. Under controlled injection it detects and attributes CPU, disk and memory faults, and has produced a correct causal explanation of a fault it was never told about. Across real history it explains roughly one incident in six, declines to explain the rest, and distinguishes “not tested” from “tested and nothing survived”.

We argue this last property matters more than the explanation rate. A tool reporting “no causal chain was supported” when the data cannot support one is more useful than a confident ranking that changes with the analysis window. In this system, getting it to say so honestly took more engineering than computing an answer did.

Future work: repeating the evaluation on second hardware, testing a fault whose cause is not the loudest metric, resolving whether the subsystem prior should admit the feedback directions it currently forbids, encrypting the database independently of EFS so that Home-edition users receive the same protection, and measuring with users whether the default view leads them to correct conclusions.

The source code, fault-injection harness and survey scripts are public at https://github.com/mathuryashash/RCA-Major_project. The collected telemetry is not released, because it records one person’s application use.

REFERENCES

- [1] V. Chandola, A. Banerjee, and V. Kumar, “Anomaly detection: A survey,” *ACM Comput. Surv.*, vol. 41, no. 3, pp. 1–58, 2009.
- [2] A. Blázquez-García, A. Conde, U. Mori, and J. A. Lozano, “A review on outlier/anomaly detection in time series data,” *ACM Comput. Surv.*, vol. 54, no. 3, pp. 1–33, 2021.
- [3] S. Hochreiter and J. Schmidhuber, “Long short-term memory,” *Neural Comput.*, vol. 9, no. 8, pp. 1735–1780, 1997.
- [4] P. Malhotra, A. Ramakrishnan, G. Anand, L. Vig, P. Agarwal, and G. Shroff, “LSTM-based encoder-decoder for multi-sensor anomaly detection,” in *Proc. ICML Anomaly Detection Workshop*, 2016.
- [5] K. Hundman, V. Constantinou, C. Laporte, I. Colwell, and T. Soderstrom, “Detecting spacecraft anomalies using LSTMs and nonparametric dynamic thresholding,” in *Proc. ACM SIGKDD*, 2018, pp. 387–395.
- [6] Y. Su, Y. Zhao, C. Niu, R. Liu, W. Sun, and D. Pei, “Robust anomaly detection for multivariate time series through stochastic recurrent neural network,” in *Proc. 25th ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, 2019, pp. 2828–2837.
- [7] J. Audibert, P. Michiardi, F. Guyard, S. Marti, and M. A. Zuluaga, “USAD: UnSupervised anomaly detection on multivariate time series,” in *Proc. 26th ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, 2020, pp. 3395–3404.
- [8] C. Zhang, D. Song, Y. Chen, X. Feng, C. Lumezanu, W. Cheng, J. Ni, B. Zong, H. Chen, and N. V. Chawla, “A deep neural network for unsupervised anomaly detection and diagnosis in multivariate time series data,” in *Proc. AAAI Conf. Artif. Intell.*, vol. 33, no. 1, 2019, pp. 1409–1416.
- [9] A. Deng and B. Hooi, “Graph neural network-based anomaly detection in multivariate time series,” in *Proc. AAAI Conf. Artif. Intell.*, vol. 35, no. 5, 2021, pp. 4027–4035.
- [10] S. Tuli, G. Casale, and N. R. Jennings, “TranAD: Deep transformer networks for anomaly detection in multivariate time series data,” *Proc. VLDB Endow.*, vol. 15, no. 6, pp. 1201–1214, 2022.
- [11] W. Xu, L. Huang, A. Fox, D. Patterson, and M. I. Jordan, “Detecting large-scale system problems by mining console logs,” in *Proc. ACM SIGOPS 22nd Symp. Operating Syst. Principles (SOSP)*, 2009, pp. 117–132.
- [12] S. He, J. Zhu, P. He, and M. R. Lyu, “Experience report: System log analysis for anomaly detection,” in *Proc. IEEE 27th Int. Symp. Softw. Rel. Eng. (ISSRE)*, 2016, pp. 207–218.
- [13] M. Du, F. Li, G. Zheng, and V. Srikumar, “DeepLog: Anomaly detection and diagnosis from system logs through deep learning,” in *Proc. ACM SIGSAC Conf. Comput. Commun. Secur. (CCS)*, 2017, pp. 1285–1298.
- [14] C. W. J. Granger, “Investigating causal relations by econometric models and cross-spectral methods,” *Econometrica*, vol. 37, no. 3, pp. 424–438, 1969.
- [15] Y. Benjamini and Y. Hochberg, “Controlling the false discovery rate: a practical and powerful approach to multiple testing,” *J. Roy. Statist. Soc. B*, vol. 57, no. 1, pp. 289–300, 1995.
- [16] A. Tank, I. Covert, N. Foti, A. Shojai, and E. B. Fox, “Neural Granger causality,” *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 8, pp. 4267–4279, 2022.
- [17] J. Runge, P. Nowack, M. Kretschmer, S. Flaxman, and D. Sejdinovic, “Detecting and quantifying causal associations in large nonlinear time series datasets,” *Sci. Adv.*, vol. 5, no. 11, Art. no. eaau4966, 2019.
- [18] M. Kim, R. Sumbaly, and S. Shah, “Root cause detection in a service-oriented architecture,” in *Proc. ACM SIGMETRICS*, 2013, pp. 93–104.
- [19] P. Chen, Y. Qi, P. Zheng, and D. Hou, “CauseInfer: automatic and distributed performance diagnosis with hierarchical causality graph in large distributed systems,” in *Proc. IEEE INFOCOM*, 2014, pp. 1887–1895.
- [20] J. Soldani and A. Brogi, “Anomaly detection and failure root cause analysis in (micro) service-based cloud applications: A survey,” *ACM Comput. Surv.*, vol. 55, no. 3, pp. 1–39, 2022.
- [21] P. Wang, J. Xu, M. Ma, W. Lin, D. Pan, Y. Wang, and P. Chen, “CloudRanger: Root cause identification for cloud native systems,” in *Proc. 18th IEEE/ACM Int. Symp. Cluster, Cloud Grid Comput. (CCGRID)*, 2018, pp. 492–502.
- [22] M. Ma, J. Xu, Y. Wang, P. Chen, Z. Zhang, and P. Wang, “AutoMAP: Diagnose your microservice-based web applications automatically,” in *Proc. Web Conf. (WWW)*, 2020, pp. 246–258.
- [23] L. Wu, J. Tordsson, E. Elmroth, and O. Kao, “MicroRCA: Root cause localization of performance issues in microservices,” in *Proc. IEEE/IFIP Netw. Oper. Manage. Symp. (NOMS)*, 2020, pp. 1–9.
- [24] Y. Meng, S. Zhang, Y. Sun, R. Zhang, Z. Hu, Y. Zhang, C. Jia, Z. Wang, and D. Pei, “Localizing failure root causes in a microservice through causality inference,” in *Proc. IEEE/ACM 28th Int. Symp. Quality of Service (IWQoS)*, 2020, pp. 1–10.
- [25] M. Li, Z. Li, K. Yin, X. Nie, W. Zhang, K. Sui, and D. Pei, “Causal inference-based root cause analysis for online service systems with intervention recognition,” in *Proc. 28th ACM SIGKDD Conf. Knowl. Discovery Data Mining*, 2022, pp. 3230–3240.
- [26] L. Pham, H. Ha, and H. Zhang, “BARO: Robust root cause analysis for microservices via multivariate Bayesian online change point detection,” *Proc. ACM Softw. Eng.*, vol. 1, no. FSE, pp. 2214–2237, 2024.
- [27] A. Ikram, S. Chakraborty, S. Mitra, S. Saini, S. Bagchi, and M. Kocaoglu, “Root cause analysis of failures in microservices through causal discovery,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 35, 2022, pp. 31 158–31 170.
- [28] L. Pham, H. Ha, and H. Zhang, “Root cause analysis for microservice system based on causal inference: How far are we?” in *Proc. 39th IEEE/ACM Int. Conf. on Automated Software Engineering (ASE)*, 2024, pp. 706–715.
- [29] D. A. Dickey and W. A. Fuller, “Distribution of the estimators for autoregressive time series with a unit root,” *J. Amer. Statist. Assoc.*, vol. 74, no. 366, pp. 427–431, 1979.
- [30] L. Page, S. Brin, R. Motwani, and T. Winograd, “The PageRank citation ranking: bringing order to the web,” Stanford InfoLab, Tech. Rep. 1999-66, 1999.